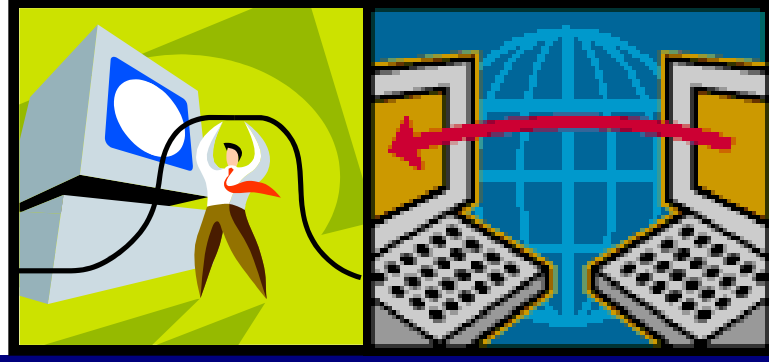

Use of ICT in Maths Education

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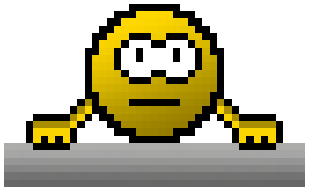
- All slides not given in this handout, and all other materials used in this lecture that are free, will be uploaded to the Blackboard after the lecture

Usual Word Problem in Algebra

- Ivy chose a two-digit number. The sum of both digits was 8. She subtracted this sum from the original number to get 27. What was the original number?
- The Mind Reader is more interesting than the usual word problems: example of using an **interactive puzzle or game** on a computer to arouse _____ in students
- Example of a **trigger activity** to introduce lecture

Objectives for Lecture



- Background: MOE's ICT _____
- **How** to use ICT to teach maths?
- **Why** use ICT to teach maths?
- **When** to use ICT?
- Planning and conducting of an ICT lesson in the computer lab

Background

- MOE's vision (1997): Thinking Schools, Learning Nation (TSLN)
- One initiative: **IT Masterplan (1997-2002)**
- IT: Information Technology
- ICT: Information and _____ Technology
- Provision of basic IT **infrastructure** and equipping teachers with **basic level** of IT integration competency

Background

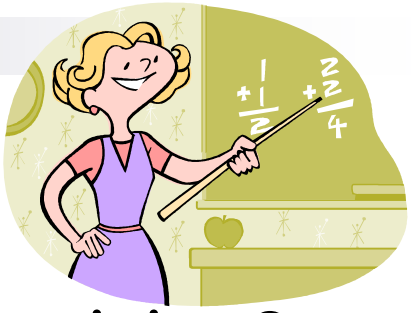
- Review: Most teachers, who used IT, used it as a presentation or demonstration tool
- **IT Masterplan II (mp2) (2003-2008)**
- Design of a more dynamic curriculum by taking into account _____ made possible by technology, and establishment of **baseline IT standards** for students
- Review: Most teachers below 30 years old are ICT savvy, but that does not mean that they can make effective use of ICT to teach[1]

[1] www.moe.gov.sg/media/speeches/2008/08/05/opening-address-by-dr-ng-eng-h-1.php

Background

- **ICT Masterplan 3 (mp3) (2009-2014)**
- They have changed the name from IT to ICT
- Vision: Harnessing ICT, Transforming Learners
- Bring ICT into the _____ of the education process by integrating ICT during the planning and design of lesson plans and **not just an add-on** to traditional ways of doing things

How to use ICT?



- So how do we use ICT to achieve this vision?
- Before invention of computers, teachers used (and are still using) _____ to show working
- So should teachers use **PowerPoint** to show step-by-step working?
- In the comics, Peter was using a **new** invention, the pencil, like the chisel, in the _____ way
- Was it effective? _____

Use PPT to show working?

- Suppose a teacher wants to use PPT to show working
- Assume you are students who have not learnt the following before: $ax + bx - 2ay - 2by$
- Main problems:
  - Students can't see how you do working _____ _____
  - Should you explain everything first (students may not follow if they don't see next line) or should you reveal next line first (cannot ask students questions)?

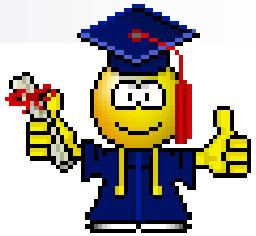
Use PPT to show working?

- What about using PPT to animate step by step?
- Main problems:
  - Using Equation Editor to type so many algebraic statements is already _____, and you want to **type each part separately to animate?**
  - Arranging the different parts in a straight line is also time-consuming
  - What happens if you animate wrongly?

Use PPT to show working?

- True story of a student teacher who used PPT to show working during practicum
- His students said **they** _____ **his working** when he asked for feedback
- Imagine what would have happened if the ST did not get feedback from his students?
- He could have wasted one whole month teaching when his students could not follow his working!

Key Lesson #1

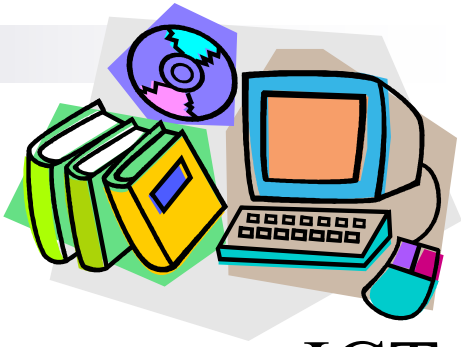


- _____ **to show working!**
- Use **whiteboard** to show working **step by step** and you can **ask questions along the way**

So what can you use PPT for?

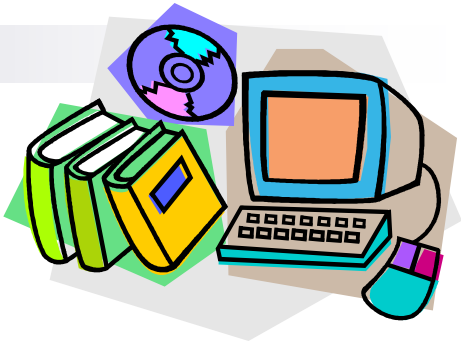
- You can use it to show _____, e.g., types of pyramids after getting students to give you some real-life examples
- These photos will be uploaded for you also

How to use ICT?



- But why did MOE spend so much money on ICT just to let you use PPT to show colourful pictures?
- You can always print out pictures using colour printer (many teachers can share one printer) and show them on a visualiser
- So what else can we use ICT for?

How to use ICT?



- We can use ICT to _____

  _____
- E.g. How do you teach students angle at centre of circle = 2 × angle at circumference?
- Do you start with a formal (deductive) proof?
- **Bruner:** _____ (concrete → pictorial → abstract)
- Students to draw circles and explore?
- Concrete but how many cases can they explore?

Geometer's Sketchpad

- Use **Geometer's Sketchpad (GSP)** templates to explore geometry
- Pedagogy: _____ **(Bruner)** using **pre-designed templates**
- **GeoGebra**: free interactive geometry software
- GSP and GeoGebra have their pros and cons
- What about exploring algebraic and calculus concepts?

LiveMath

- **Computer Algebra System (CAS)**, e.g. Maple, used traditionally by mathematicians, scientists and engineers to solve complicated algebraic and calculus problems at higher levels
- But students still need to perform algebraic and calculus operations (such as factorisation and differentiation) using paper and pencil
- So _____ **students to use CAS to perform algebraic and calculus manipulations**
- Use **interactive** CAS (e.g., LiveMath) to **explore algebraic and calculus concepts**

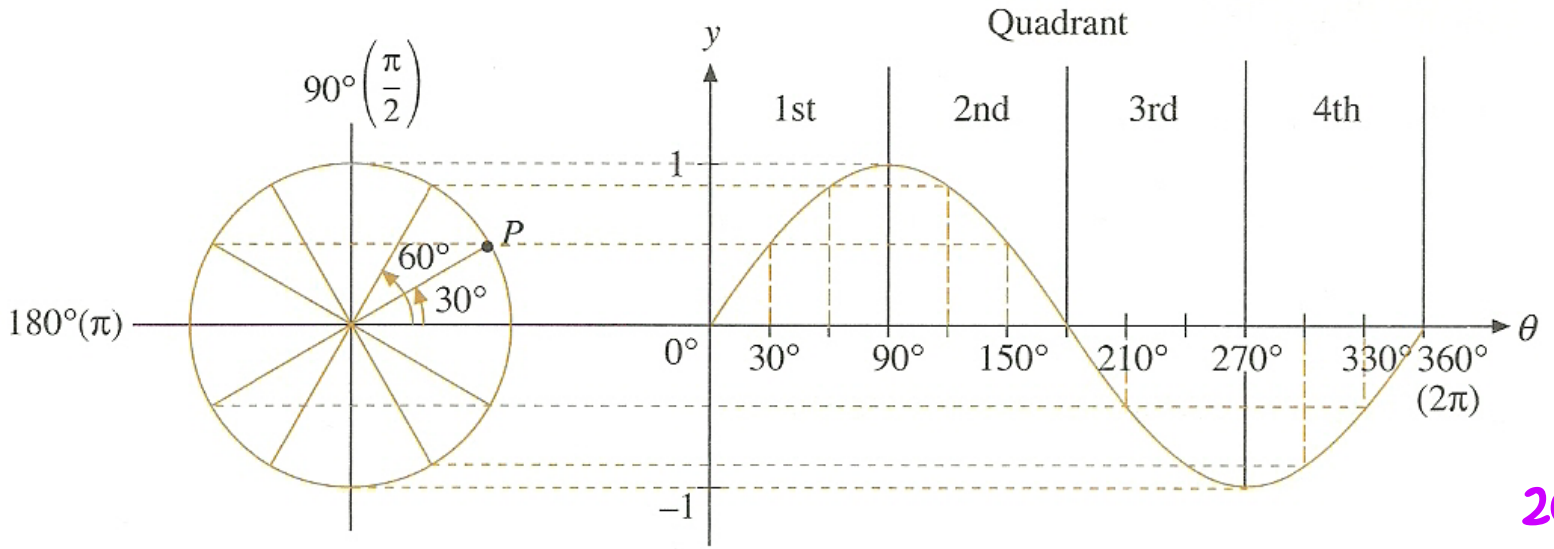
Pre-designed Templates

- Should you use **pre-designed templates** or teach students to design the templates themselves?
- If you let students design templates, they will spend more time on the technical aspects than _____ **itself**
- So you should give them the templates to explore
- But you can still ask them to carry out **simple operations** such as measuring angle or using in-built calculator to perform some calculations
- Provide **worksheet** for students to record finding

Q1. Do you prefer teacher demo or student hands-on? Why?

Hands-on vs. Demo

- Teachers attending ICT workshop prefer _____ _____ but they want to _____ to their students?!
- But is there a place for demo?
- E.g., how do you show students the use of a unit circle to generate a sine curve?
- Textbook shows this static diagram:



ICT as Visual Aids

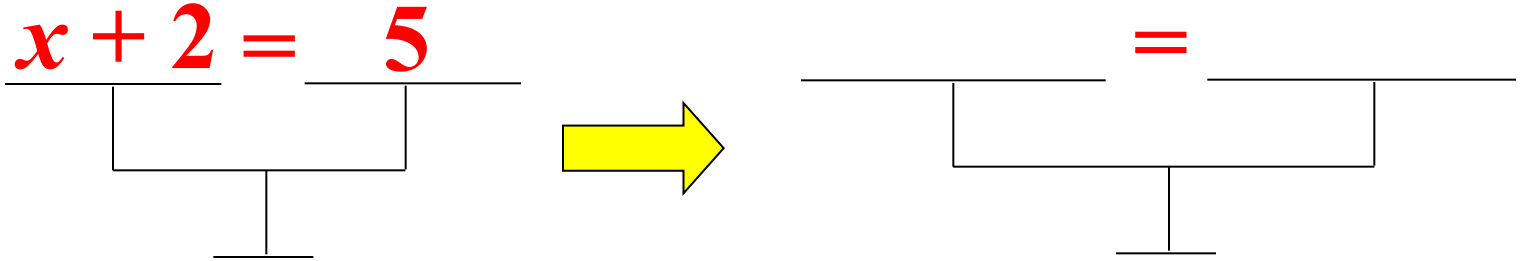
- After explaining while demonstrating using GSP, you are done!
- So there is a place for demo when content is too difficult for students to explore
- But whenever possible, let students have hands-on so that they can learn better
- In this example, ICT is used as a **visual aid** to **help students to visualise better**

ICT as _____ Aids

- E.g., you need to show students how to press calculator when teaching them how to evaluate expressions like sin 30°, $\cos^{-1} 0.5$, ln 7.16, etc.
- You can use an **emulator**

ICT as Virtual Manipulatives

- You may have learnt how to use a balance to teach the solving of a linear equation:

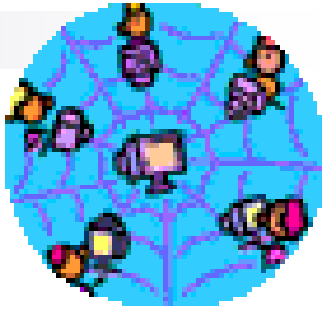


- Quite static, so you can use a real balance to demonstrate
- But what happens if the equation is $2x - 5 = -3$? How are you going to show – 5 and –3?
- http://nlvm.usu.edu/en/nav/category_g_3_t_2.html (National Library of Virtual Manipulatives)

ICT for e-learning

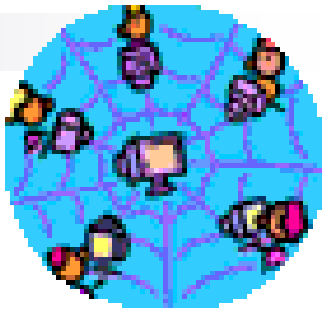
- www.math.com/school/subject2/lessons/S2U3L1DP.html
- Quite _____: how is this different from just using textbooks?
- To make it more interactive, many technology experts, who are not educators, will suggest the use of discussion forums, teleconferences, blogs, wikis, etc.
- But these are the hardware (or software)
- What about the **content**? How to make the content more **interactive**?

ICT for e-learning



- **JavaSketchpad** (Web-based version of GSP)
- This is how you can **make the maths content more** _____
- End users (students) don't need GSP software to interact with the templates
- GeoGebra and LiveMath also have Web-based versions: end users don't need the software but they just need to install a free software called a plug-in

Resources from Internet



- Song: Search “I will _____” in YouTube
- Video: Search “____ divided by ___” in YouTube

ICT to help Teachers

- Use of ICT for students vs. use of ICT for teachers
- E.g., use of Equation Editor and MS Draw to prepare worksheets, test and exam papers
- Free software have their limitations
- How do you shade the following?

ICT to help Teachers

- **FXDraw3**: not free; from Australia; **by teachers for teachers**, so they know what you want!
- Gallery contains many pre-drawn figures
- Lines and curves may not look smooth on computer screen, but they will look good when printed out, just like in your lecture notes

Summary: 8 Ways to use ICT

- Interactive **puzzles** or games, **songs** and **videos** to interest students
- Show colourful pictures
- _____ **mathematical concepts**
- **Visual aids** to help students visualise
- Teaching aids, e.g., emulator
- **Virtual manipulatives**
- E-learning
- Prepare worksheets, test and exam papers

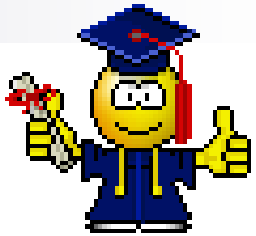
Why use ICT?

- **Bruner: CPA** (concrete → pictorial → abstract)
- "The computer … has the **ability to make the abstract concrete**." (Turkle & Papert, 1990)
- So ICT can enhance student learning by _____ **the gap** between abstract concepts and concrete experiences
  - Explore mathematical concepts
  - Visual aids to help students visualise
  - Virtual manipulatives

So when to use ICT?

- When it **helps** to _____
  - If it does not help, don't use!
  - E.g., don't use PowerPoint to show working
  - E.g., sometimes concrete manipulatives are better than virtual manipulatives, so don't use virtual ones
- _____ use of ICT in maths education
- No time and **not possible** to guide students to discover everything
- **Balance: use different modes** of teaching

Key Lesson #2



- Use ICT only when it _____ _____ , e.g., by making the abstract more concrete

Planning an ICT lesson

- First time is always the hardest
- Visit comp lab a few days before: see what are available and **test** teacher computer and a few _____ computers
- Visit comp lab the day or the lesson before: check equipment **still works**
- Try out template and worksheet _____: you may discover what don't work and you need to fix problems before lesson

Conducting an ICT lesson

- Gather students at the front of comp lab. Why?
- Explain and demo what you want them to do
- _____:

  time consuming, boring and what is left for students to explore?
- **Plan** what you need to demo, esp. the **difficult parts**, or else it's hard to gather students again

Conducting an ICT lesson

- Break **two-period lesson** into _____: demo → hands-on → demo → hands-on
  - If too many segments, hard to gather students
  - If only one segment, hard to demo second part of lesson if students don't understand first part
- Walk around to **monitor progress** and to keep students **on task**

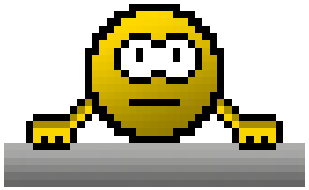
Concluding an ICT lesson

- Gather them again for **consolidation** and conclusion
- Need to check that students have **discovered the correct things**

Key Lesson #3

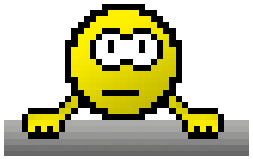
- You _____ **the computer lab** at least a few days before your first ICT lesson to make sure everything works

Objectives for Lecture



- Background: MOE's ICT Masterplans
- **How** to use ICT to teach maths?
- **Why** use ICT to teach maths?
- **When** to use ICT?
- Planning and conducting of an ICT lesson in the computer lab
- This lecture will be **complemented by the computer lab sessions in your tutorials**

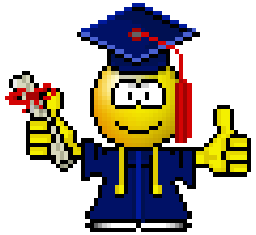
Q2: Reflection



- Will you use ICT to teach maths? Why or why not?

Conclusion

- One key message: _____
- Your duty is not to teach *per se*, but to teach so that **student learning** takes place
- _____ use of ICT to **enhance** student learning



- **Don't use PPT to show step-by-step working**